서강대학교
SOGANG UNIVERSITY

Physical Electronics Ⅱ
Semiconductor Electronics Ⅱ

TCAD simulation : MOSFET

Content

1. Introduction
2. Long channel MOSFET
3. Body effect
4. Non-ideal effects
5. Short channel effects
6. Summary

Content

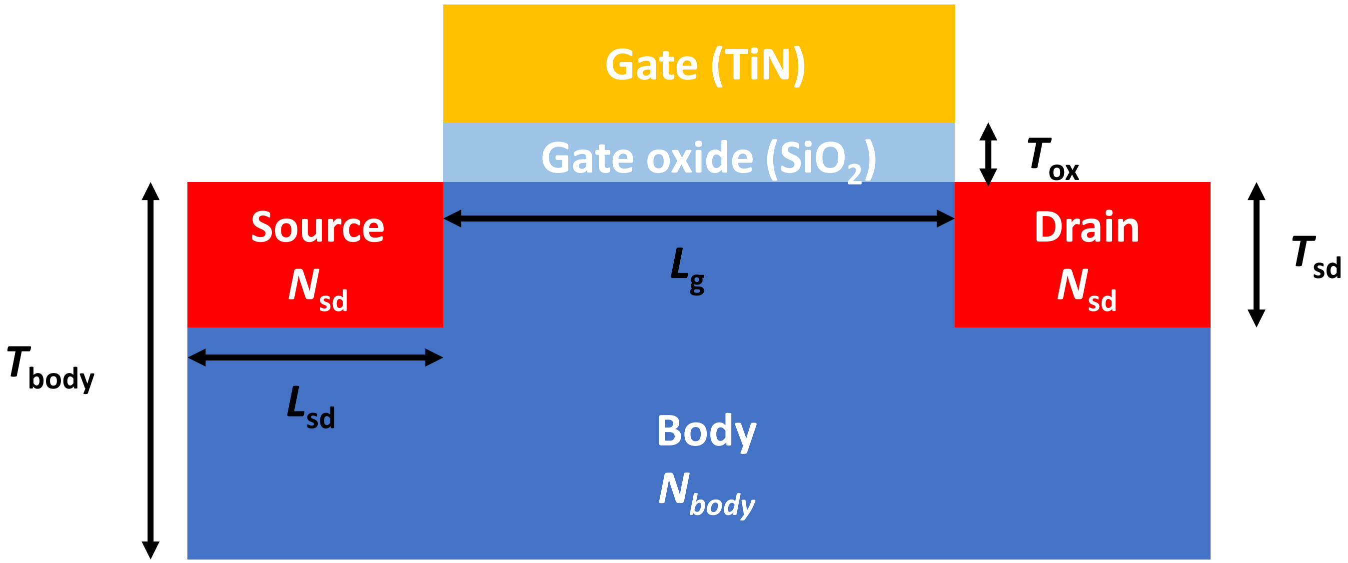
Reference

- ✓ 1 μm 공정은 1980년대 쓰이던 공정 기술
- ✓ 이시기에 상용화된 제품 중 **Intel386™ SX MICROPROCESSOR** data sheet 참고

V_{CC}	I	8-10,21,32,39 42,48,57,69, 71,84,91,97	**System Power** provides the +5V nominal DC supply input.

- ✓ 동작 전압 5.0 V로 설정하여 시뮬레이션 진행

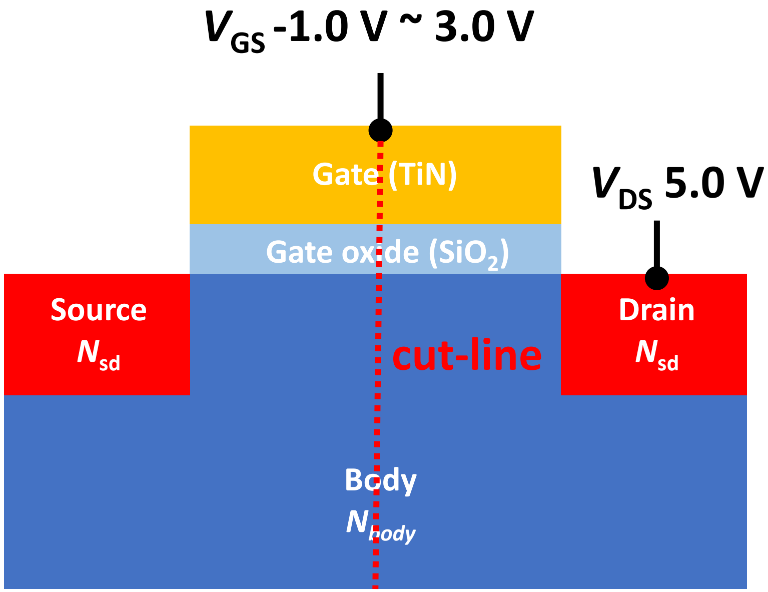
Device structure



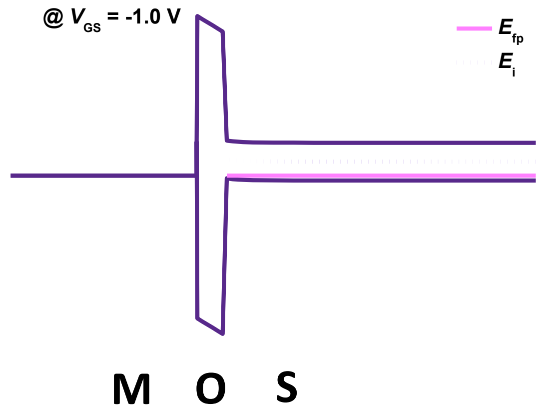
Parameters	Values
L_g	1 µm
L_{sd}	0.5 µm
T_{body}	1.2 µm
T_{sd}	0.3 µm
T_{ox}	0.02 µm
N_{body}	Boron 10^{17} cm^{-3}
N_{sd}	Arsenic 10^{20} cm^{-3}
S/D Gaussian factor	0.35

Content

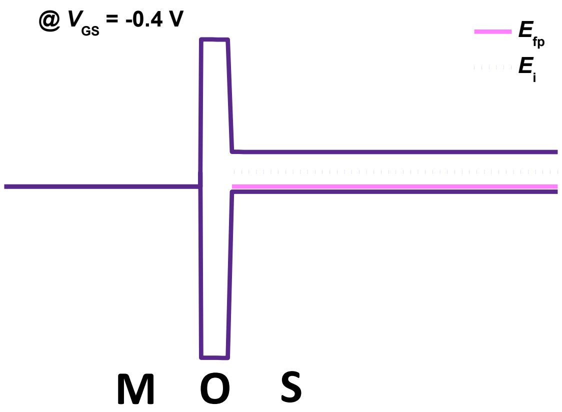
Operation region



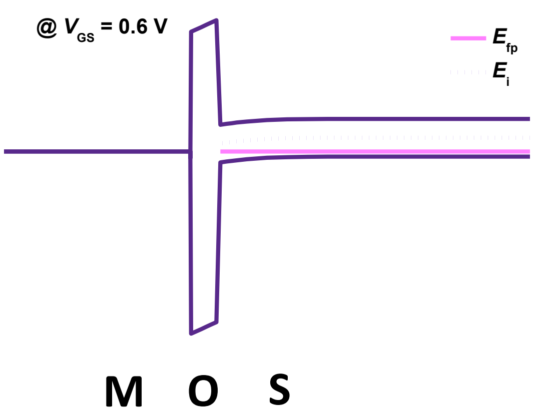
Accumulation



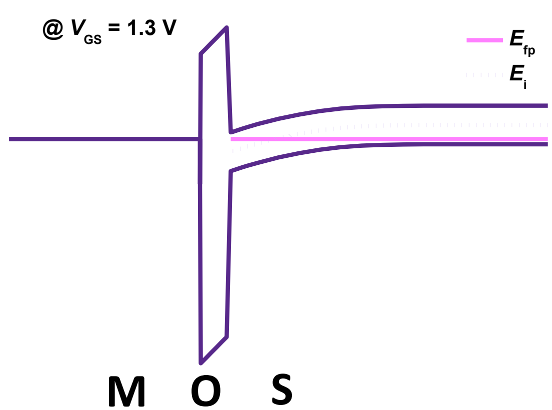
Flat band



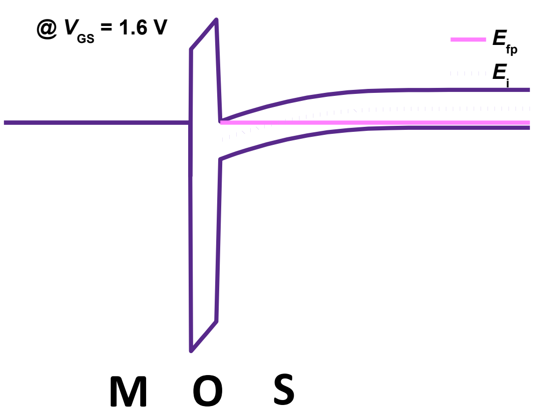
Depletion



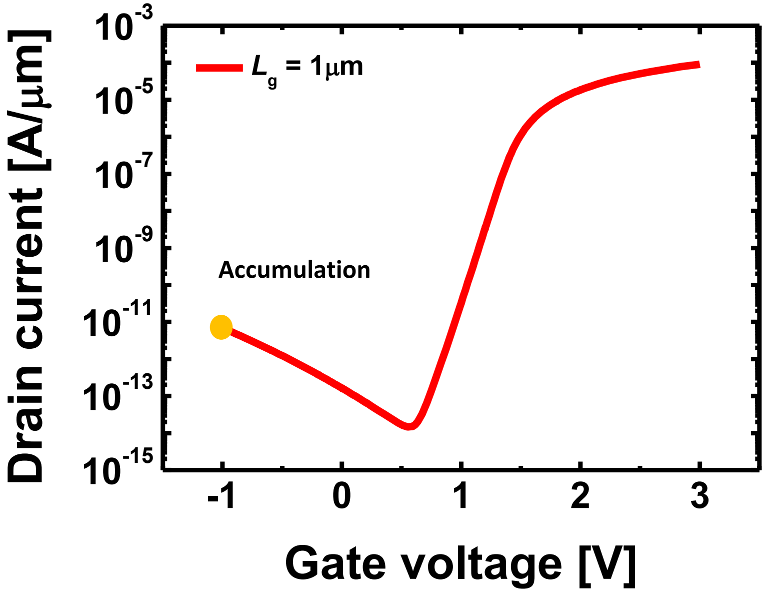
Threshold

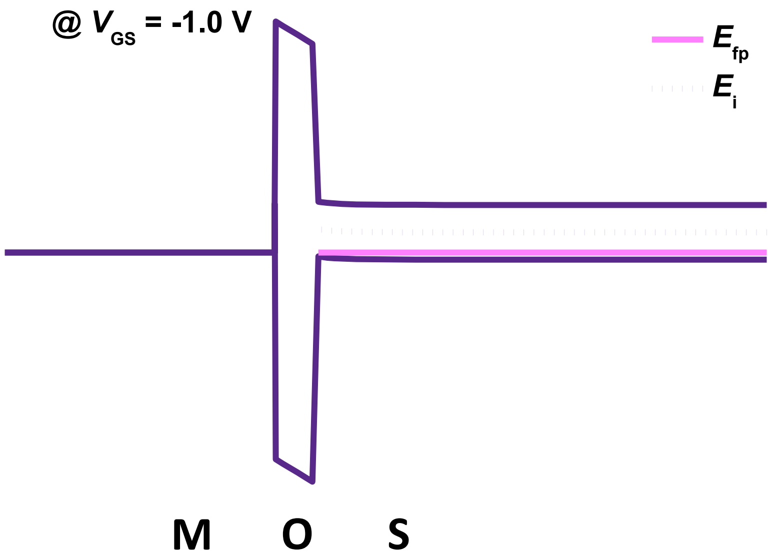


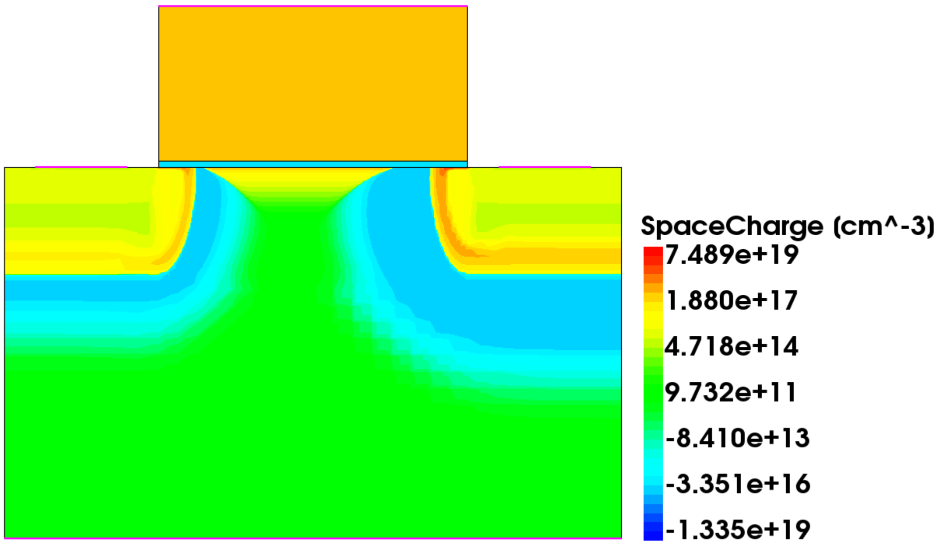
Strong inversion

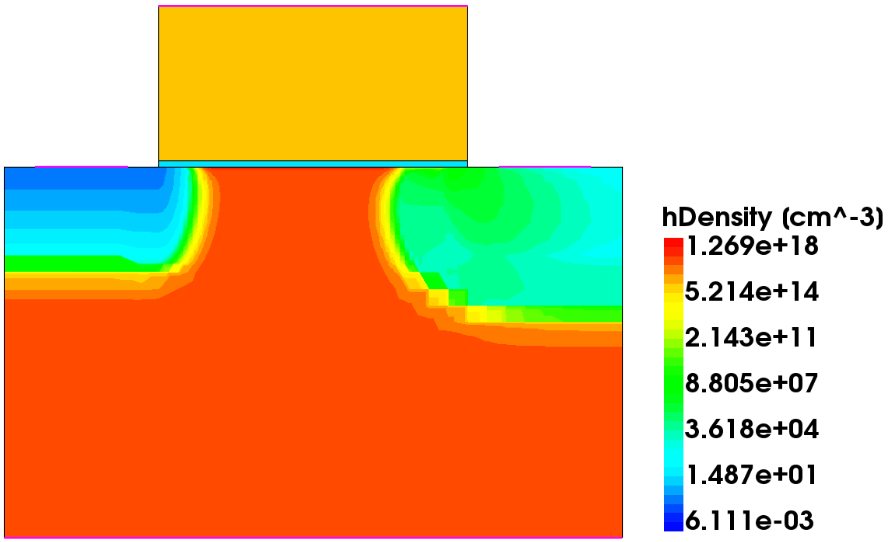


Operation region (Accumulation)

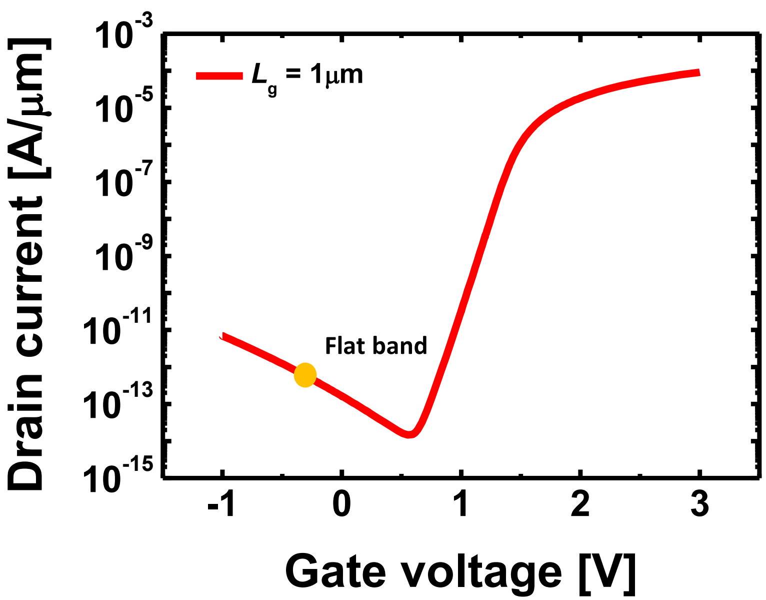


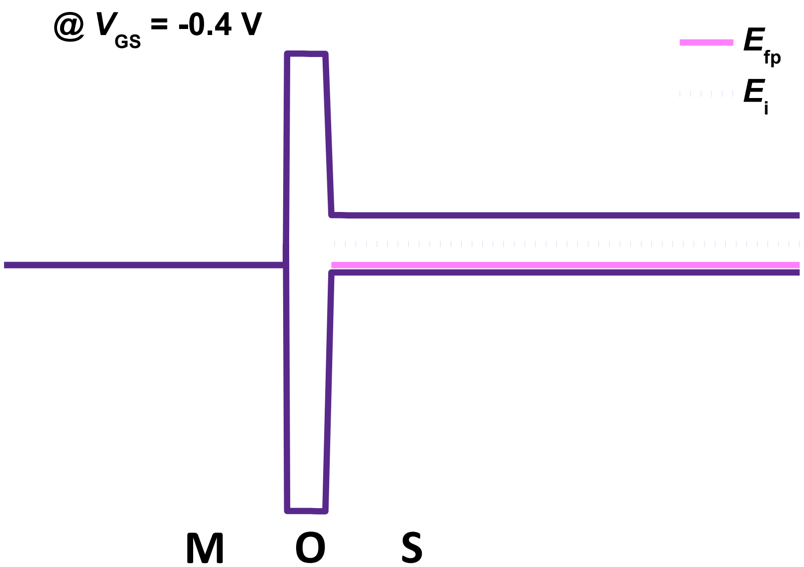


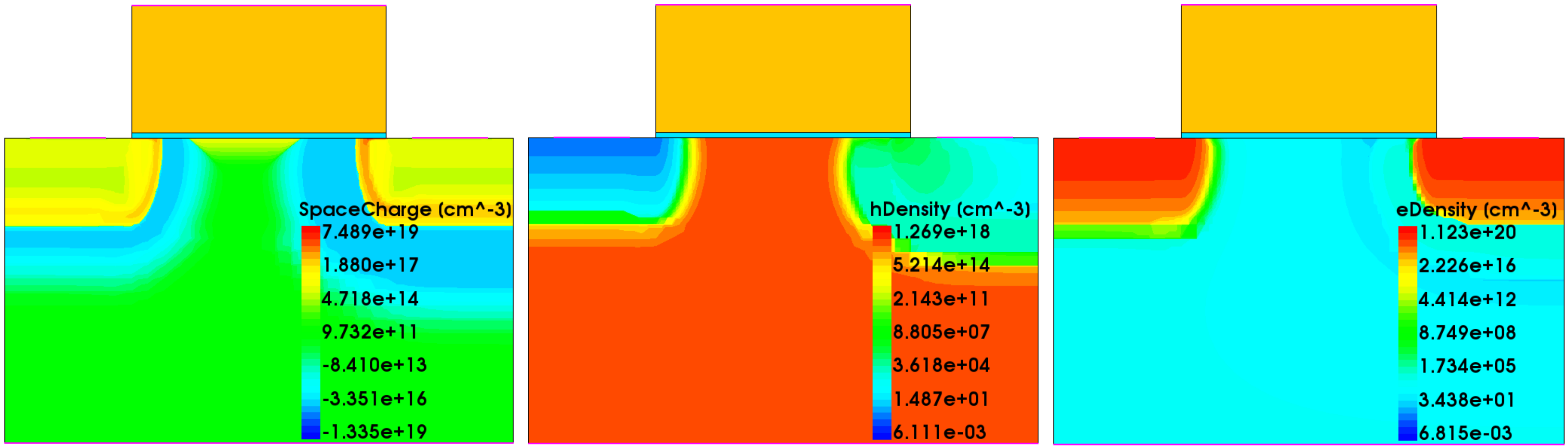




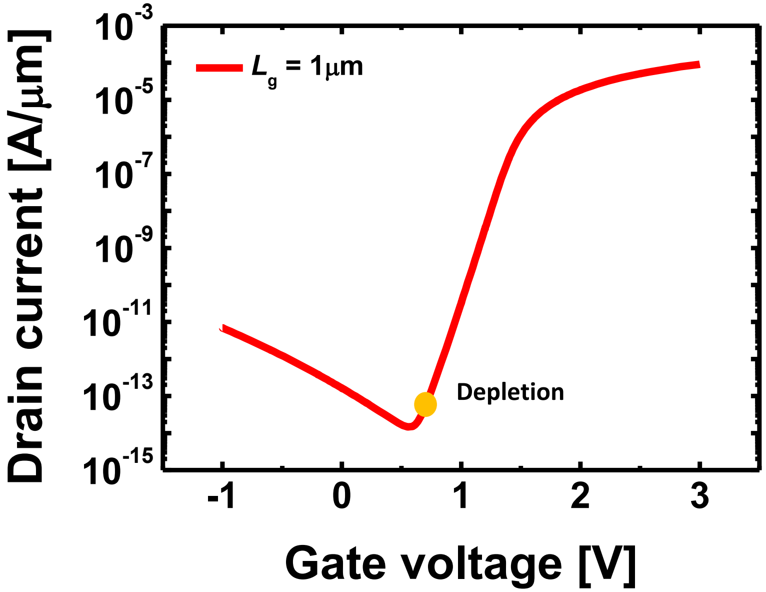
Operation region (Flat band)

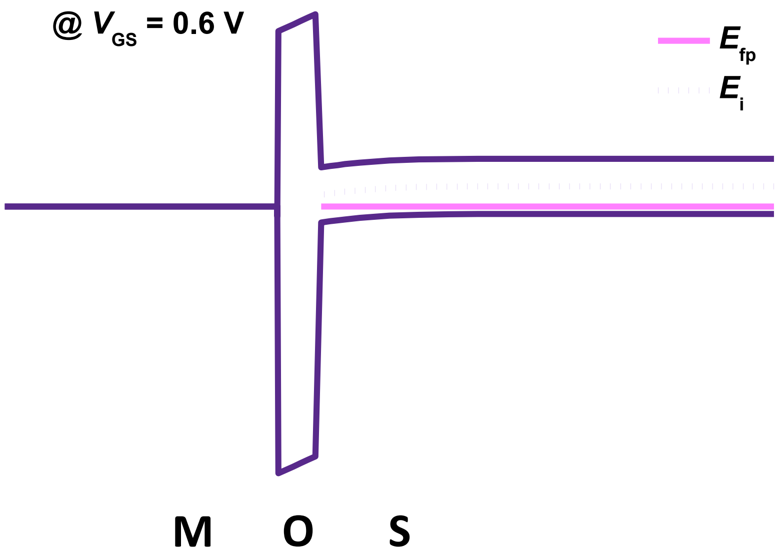


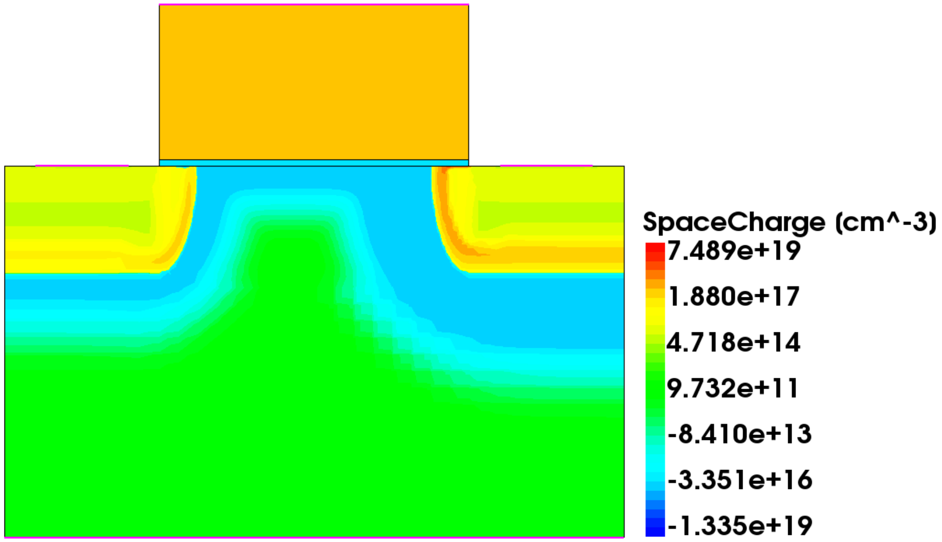


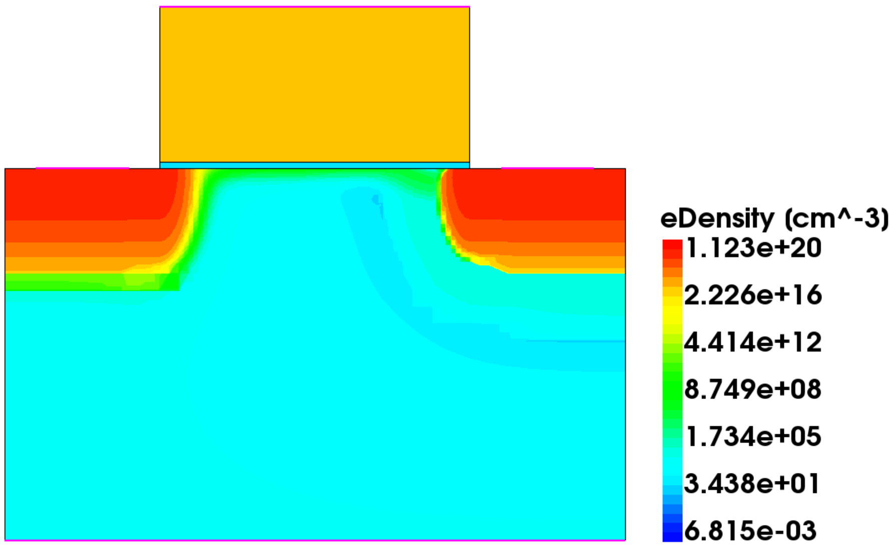


Operation region (Depletion)

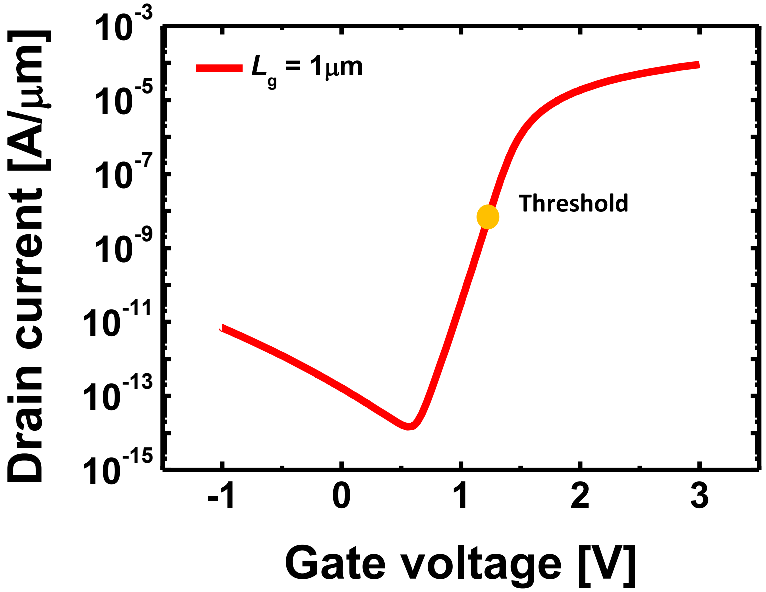


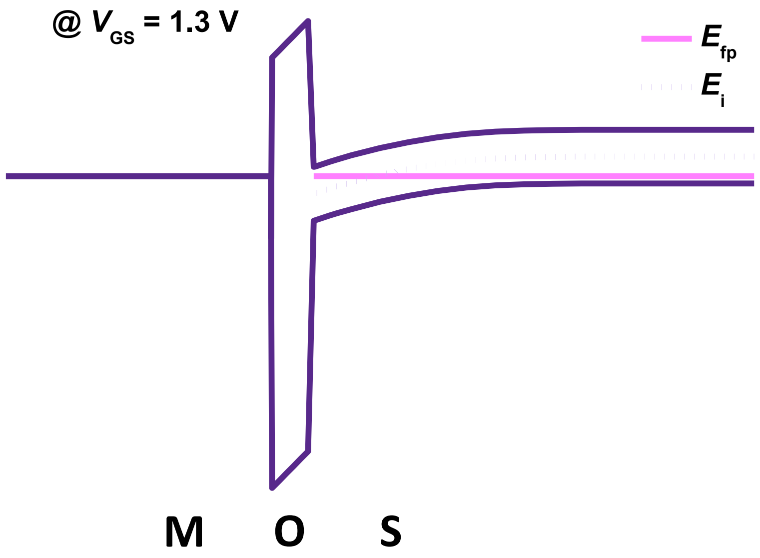


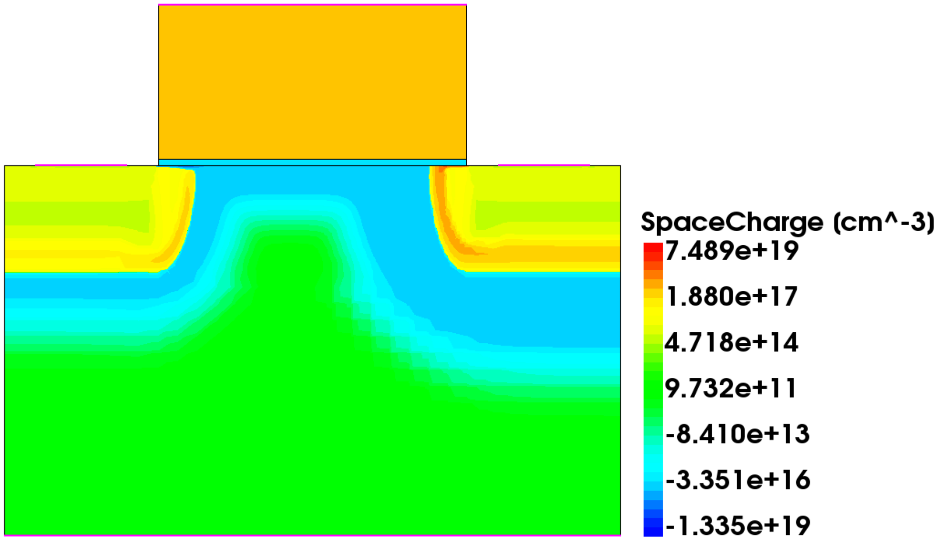


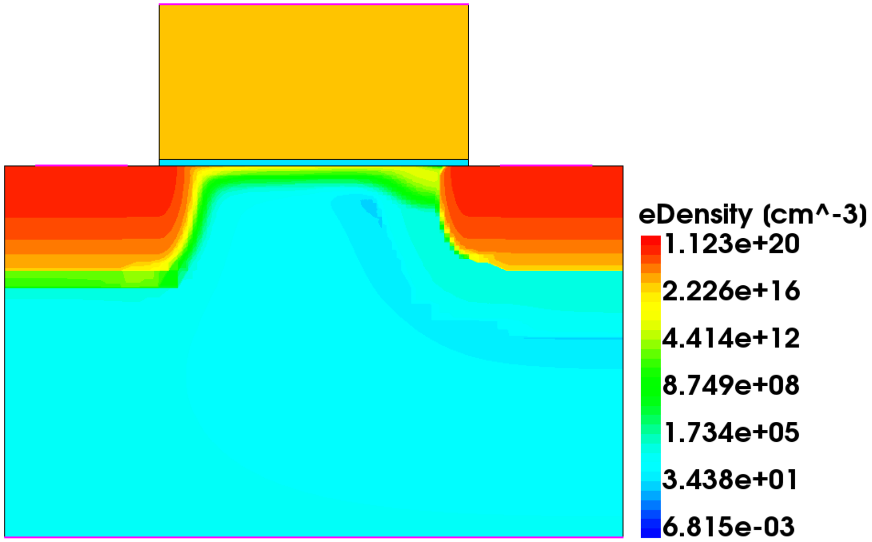


Operation region (Threshold)

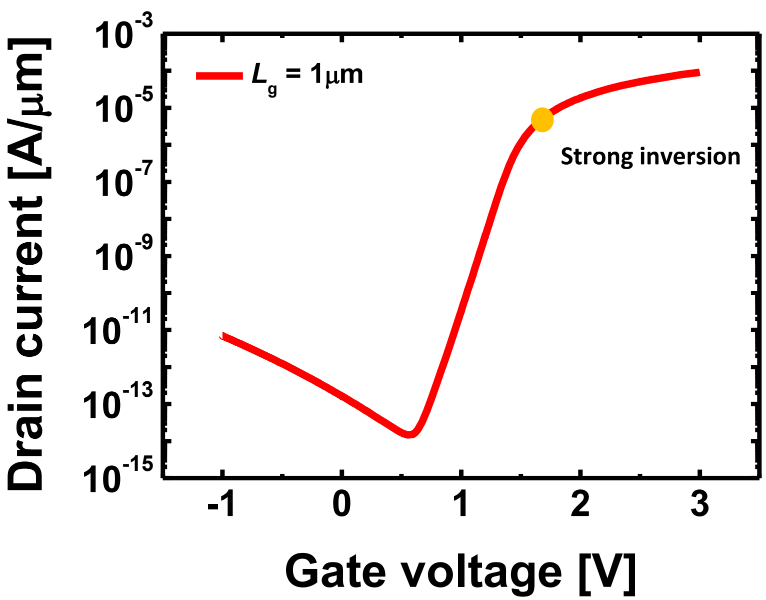


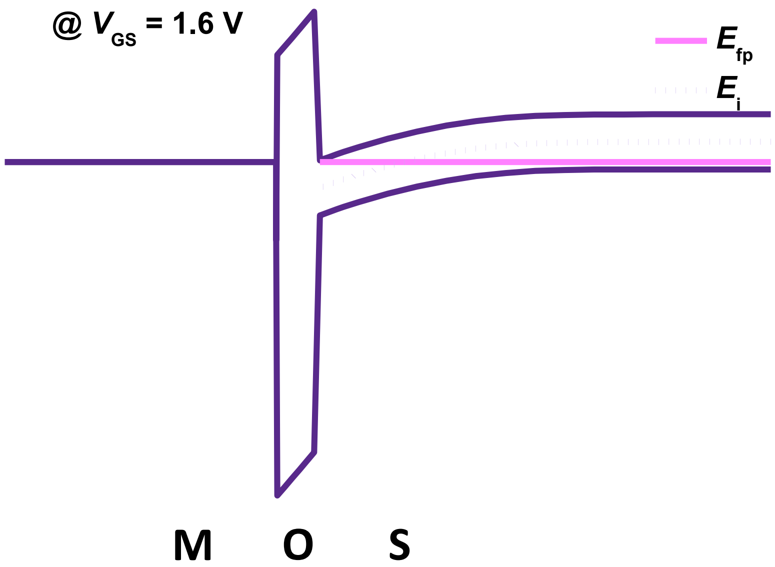


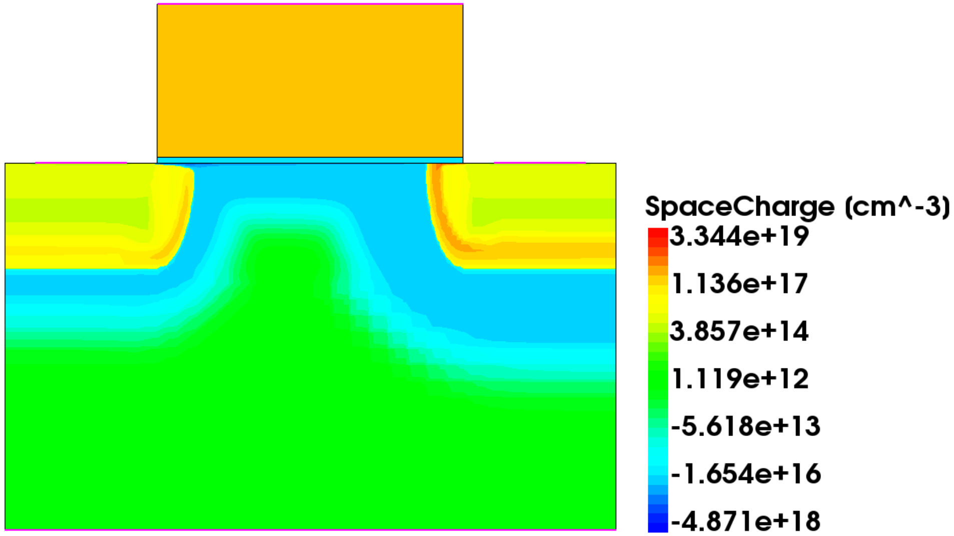


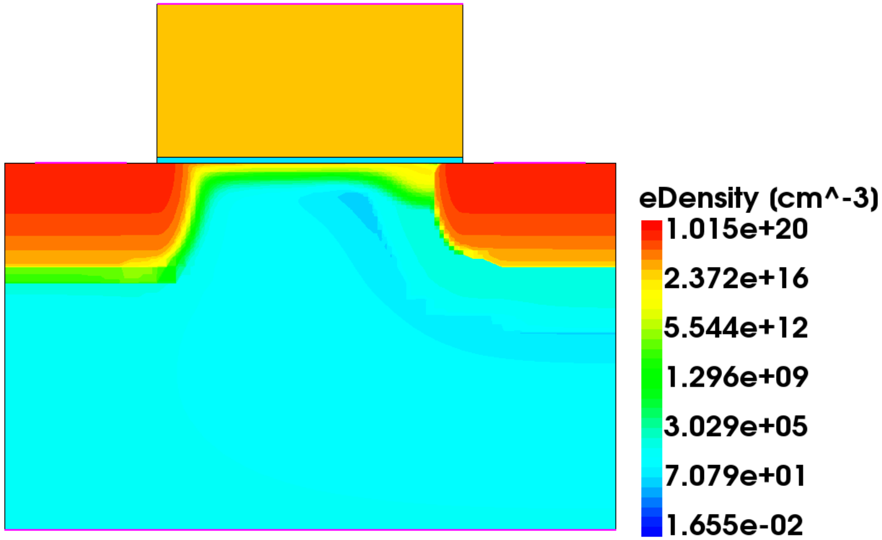


Operation region (Strong inversion)





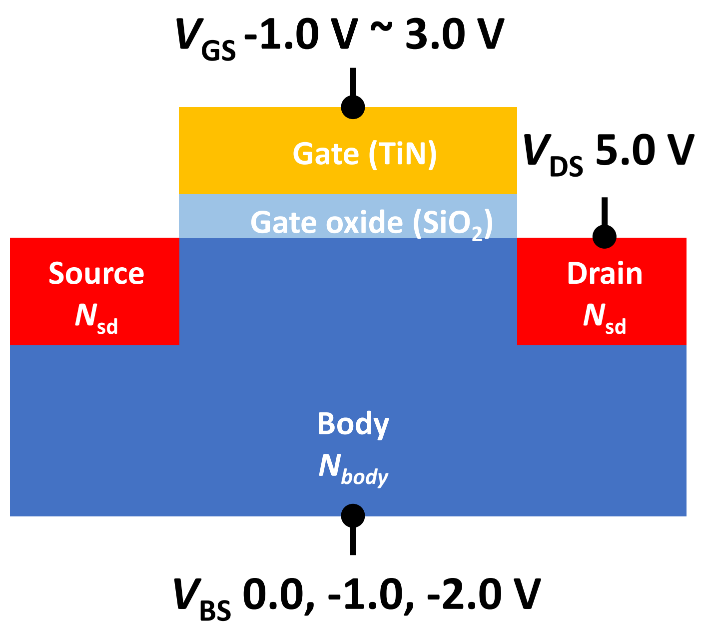


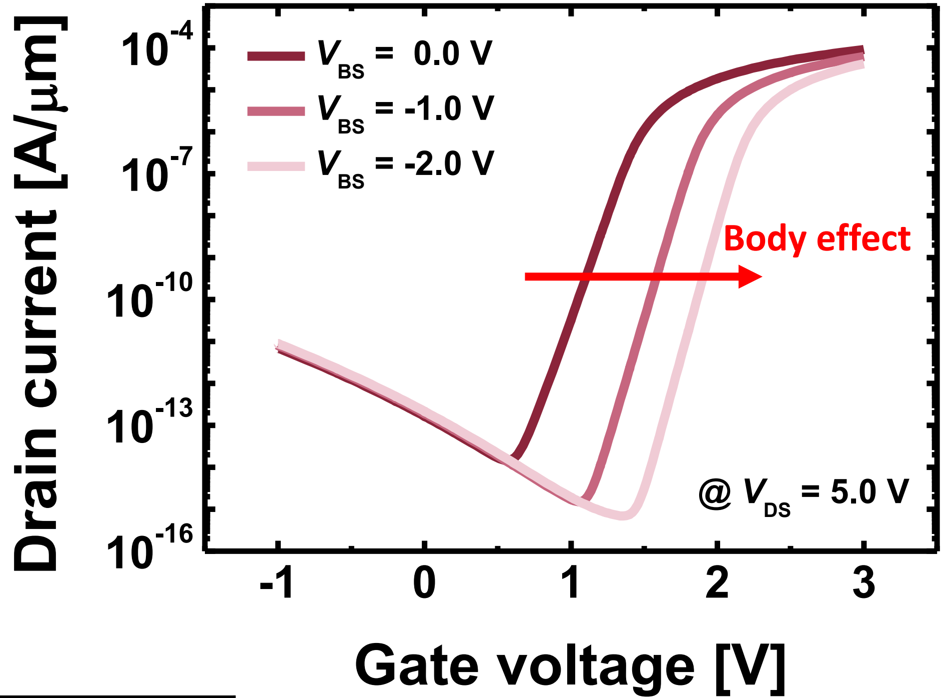


Content

Body effect

L_g = 1.00 μm



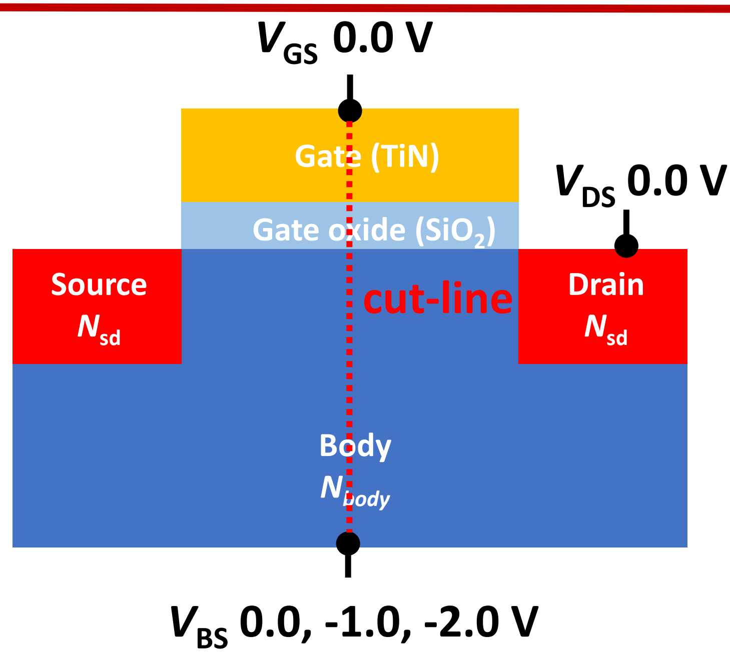


V_{BS} [V]	V_{th} [V]
0.0	1.37
-1.0	1.81
-2.0	2.12

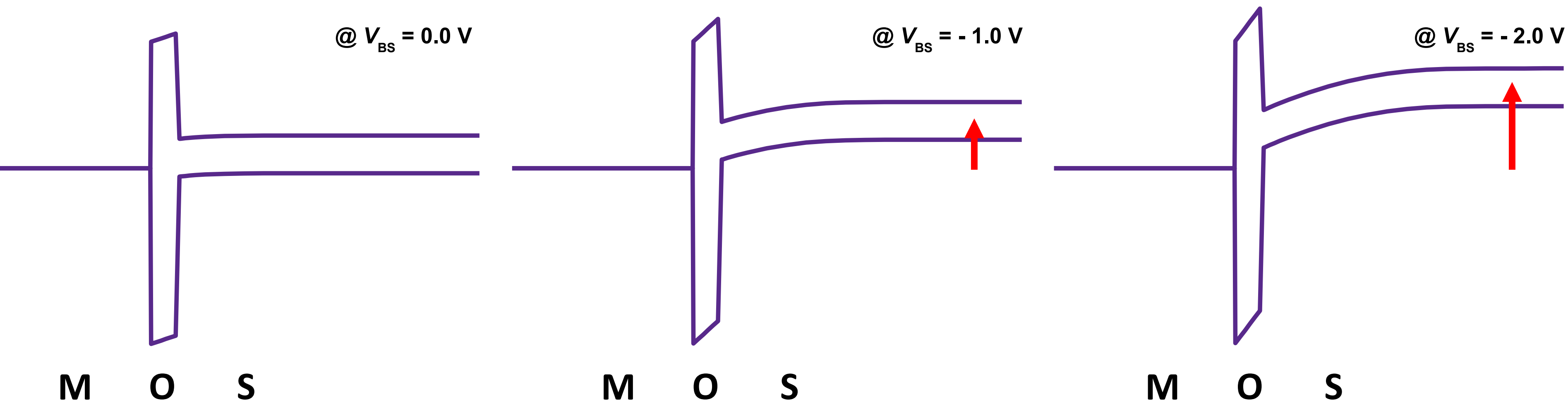
V_{th} 증가

✓ Inspect tool 활용하여 V_{th} 추출

Body effect



V_{GS} 0.0 V
Gate (TiN)
Gate oxide (SiO_2)
V_{DS} 0.0 V
Source
N_{sd}
cut-line
Drain
N_{sd}
Body
N_{body}
V_{BS} 0.0, -1.0, -2.0 V



@ V_{BS} = 0.0 V
M O S
@ V_{BS} = - 1.0 V
M O S
@ V_{BS} = - 2.0 V
M O S

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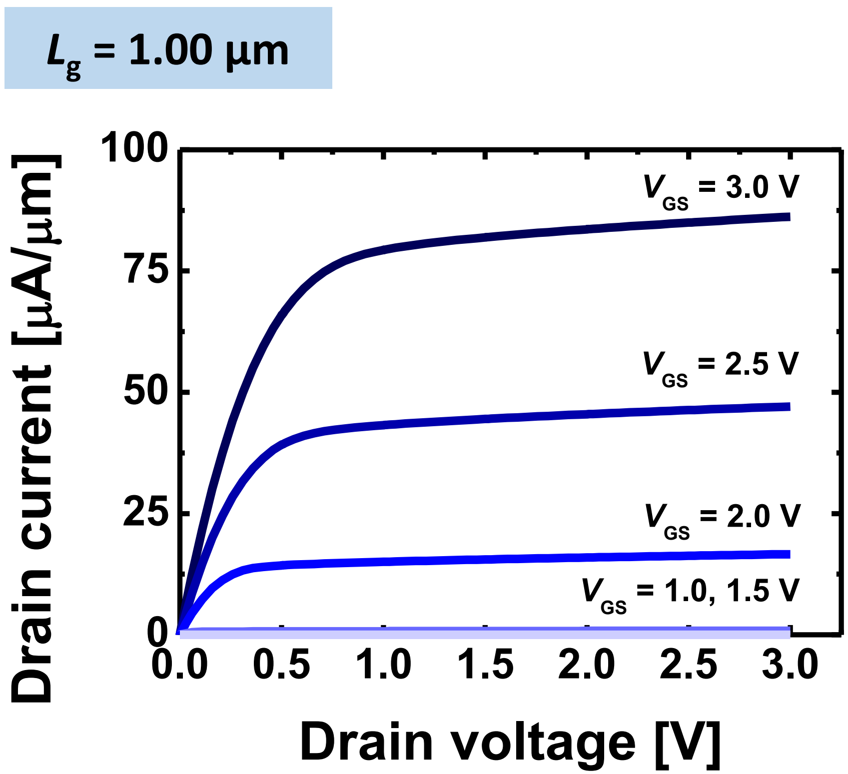
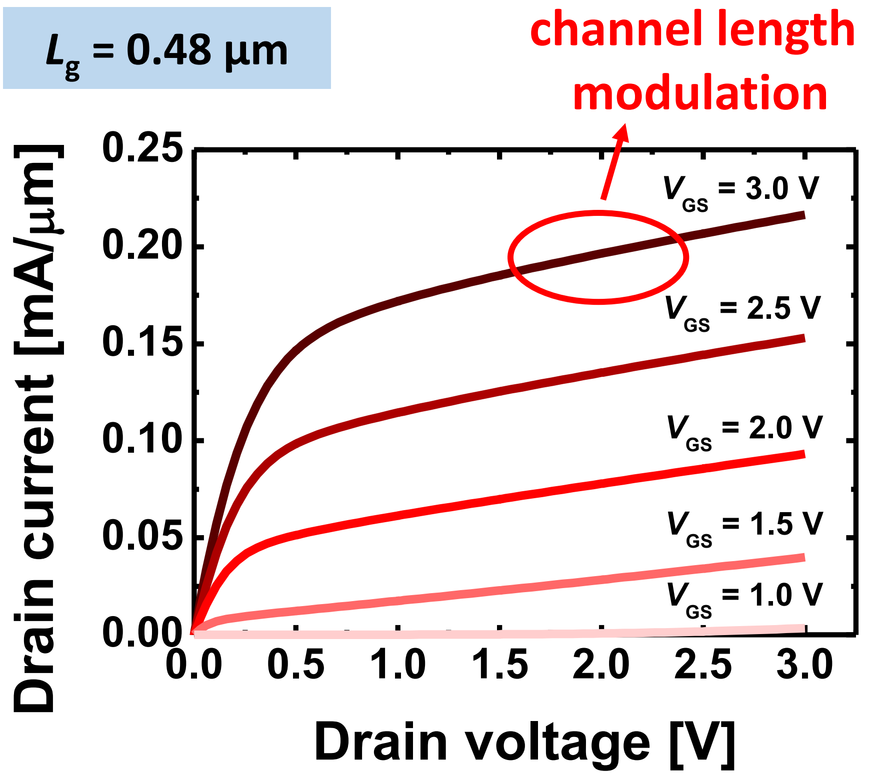
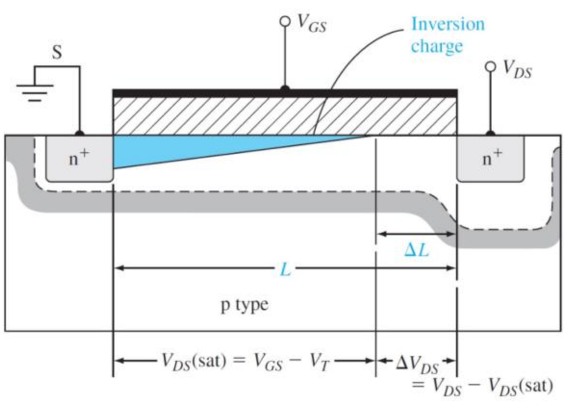
Channel length modulation





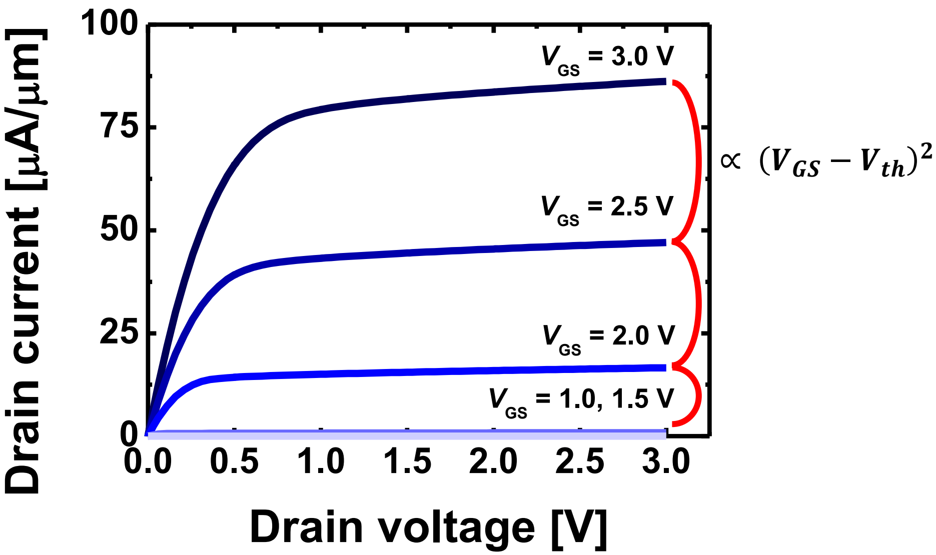


$$I_{D,sat} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{th})^2$$

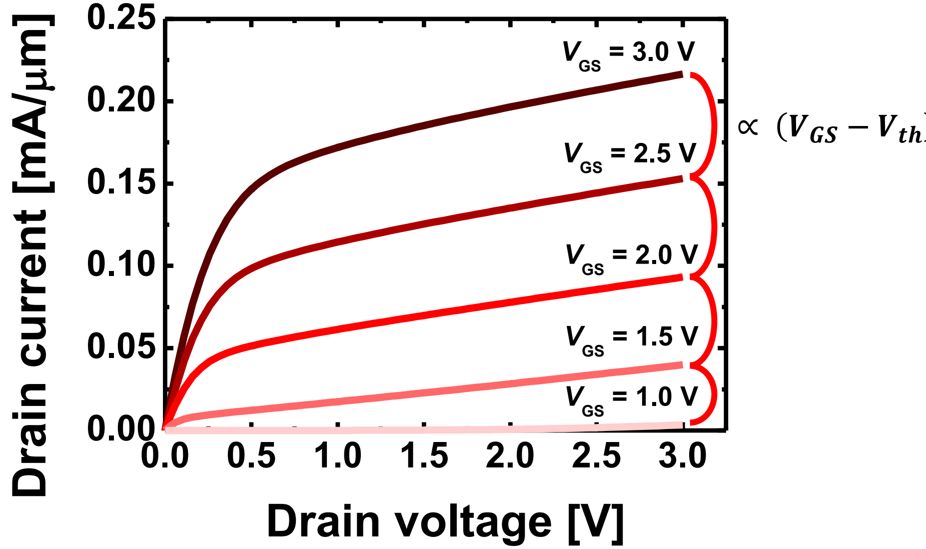
$$\therefore I'_D = \left(\frac{L}{L-\Delta L}\right) I_{D,sat}$$

Velocity saturation

L_g = 1.00 μm



L_g = 0.48 μm



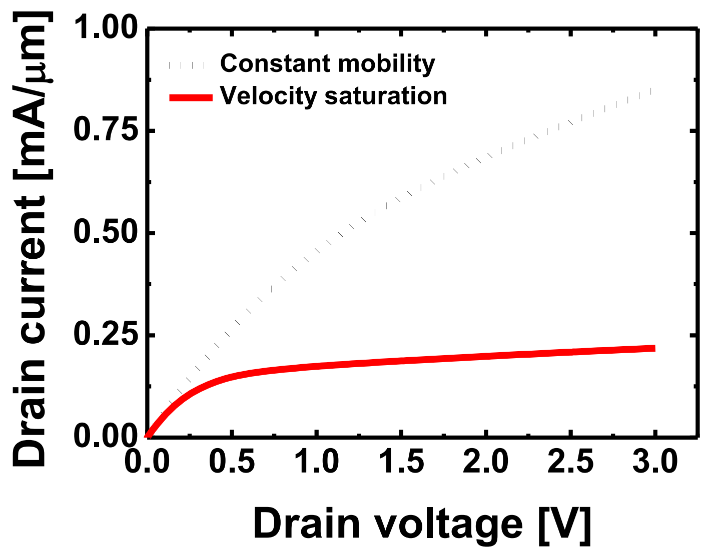
✓ **Long channel**

$$I_{d,sat} = \frac{1}{2}\mu C_{ox}\frac{W}{L}(V_{GS} - V_{th})^2$$

✓ **Short channel**

$$I_{d,sat} = C_{ox}W(V_{GS} - V_{th})v_{sat}$$

❖ **Physics의 mobility (HighFieldSaturation) 적용시 Velocity saturation 구현 가능**



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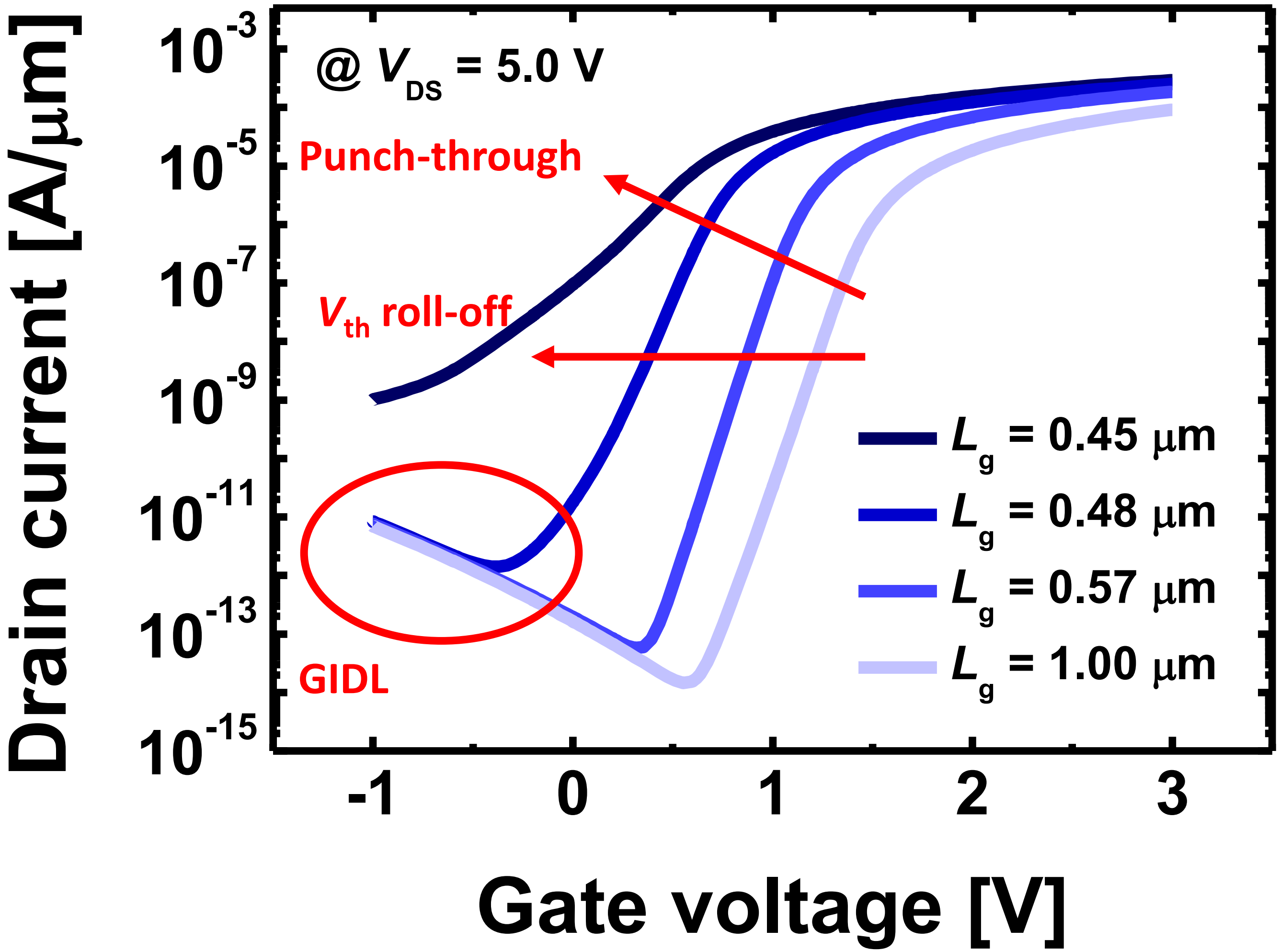
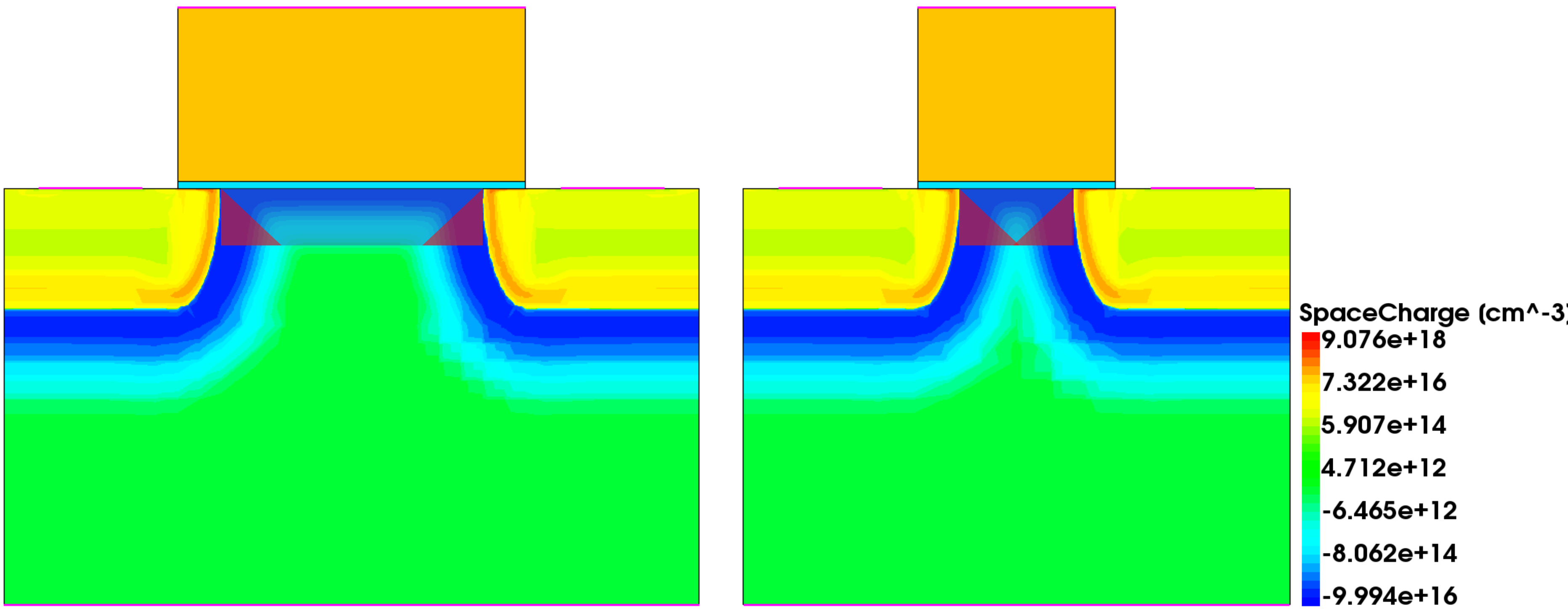
I_D-V_G

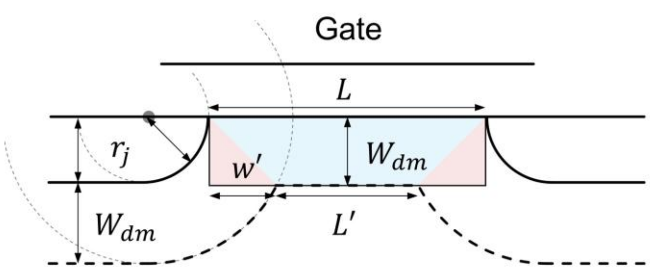


Drain current [A/μm]
10^-3
10^-5
10^-7
10^-9
10^-11
10^-13
10^-15
@ V_DS = 5.0 V
Punch-through
V_th roll-off
GIDL
L_g = 0.45 μm
L_g = 0.48 μm
L_g = 0.57 μm
L_g = 1.00 μm
-1
0
1
2
3
Gate voltage [V]

V_{th} roll-off

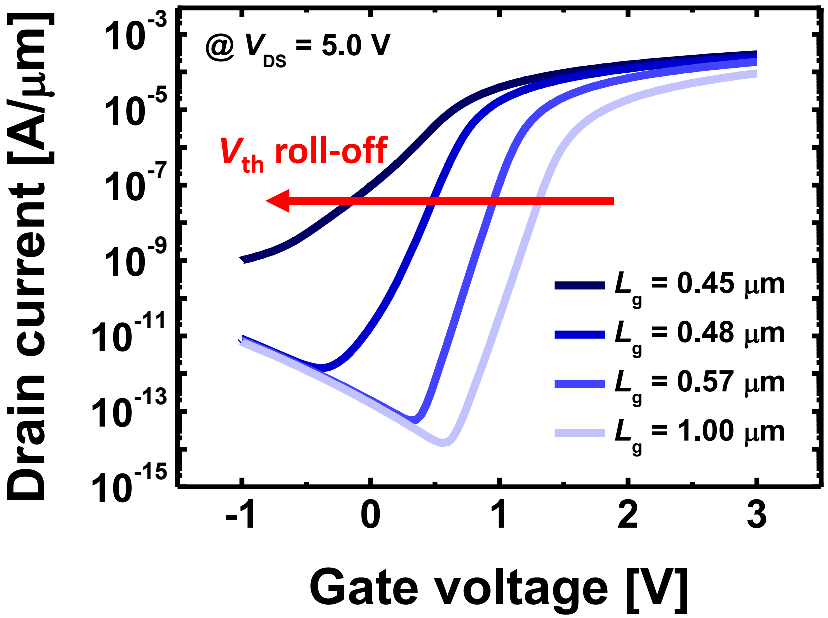
L_g = 1.00 µm | **L_g = 0.57 µm**

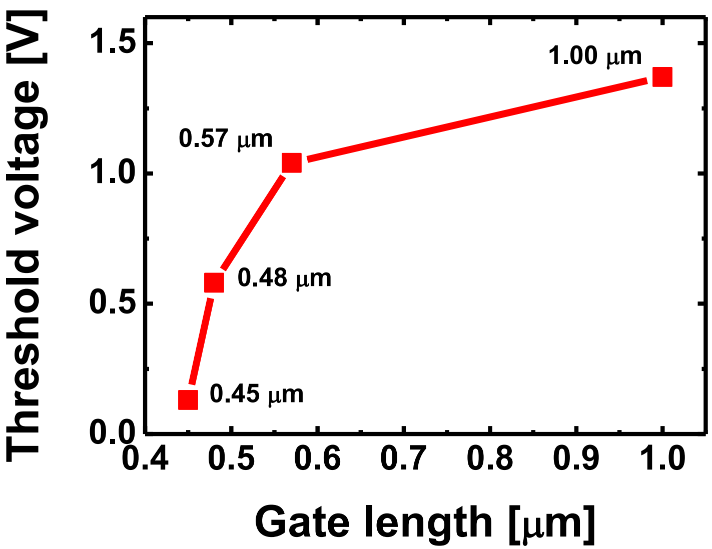




$$\Delta V_{th} = -\frac{qN_a W_{dm}}{C_{ox}}\left(\frac{r_j}{L}\right)\left(\sqrt{1+2W_{dm}/r_j}-1\right)$$

V_{th} roll-off





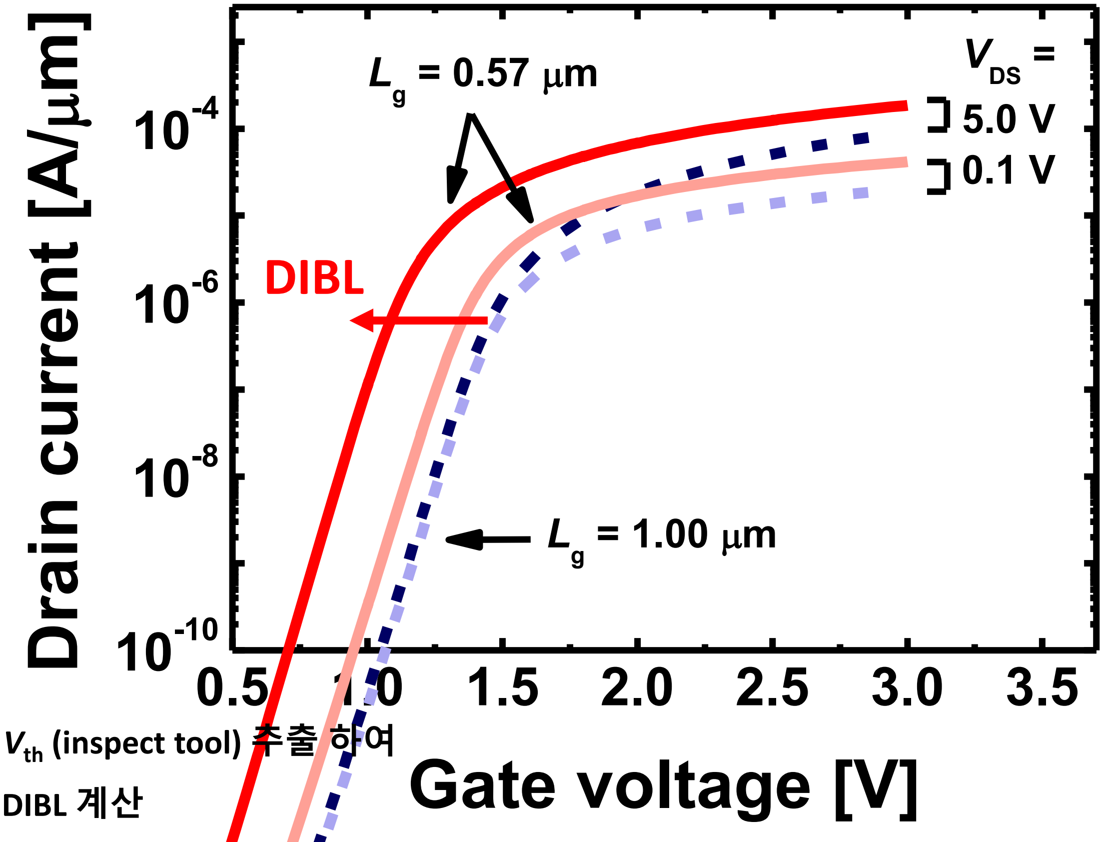
L_g [μm]	V_{th} [V]
1.00	1.37
0.57	1.04
0.48	0.58
0.45	0.13

❖ V_{th}는 $I_D = 10^{-7}$ A/μm x (W/L) [A/μm] 에서 추출

(강의 자료 W10-2 p.366 참고)

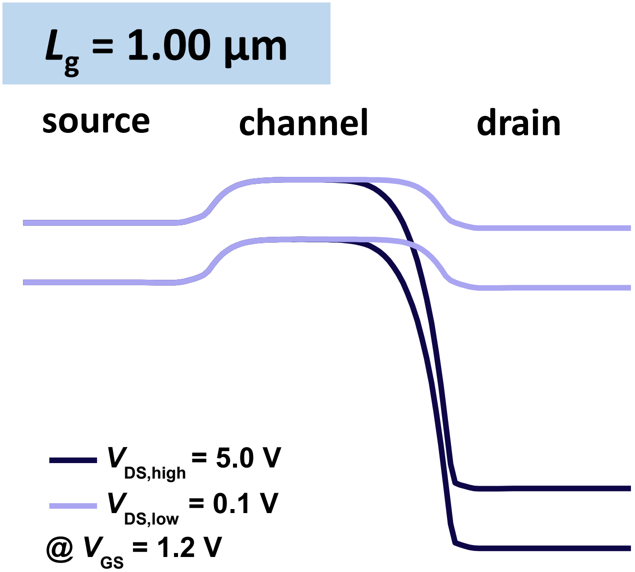
DIBL

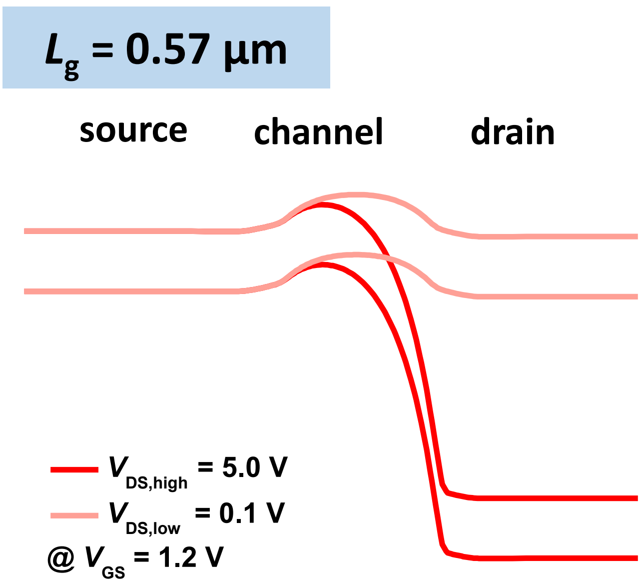
✓ High V_{DS} = 5.0 V, Low V_{DS} = 0.1 V



* V_{th} (inspect tool) 추출 하여
DIBL 계산

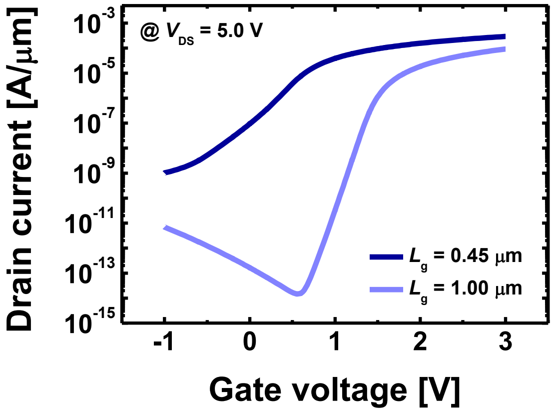
L_g [μm]	DIBL [mV/V]
1.00	4.49
0.57	53.27

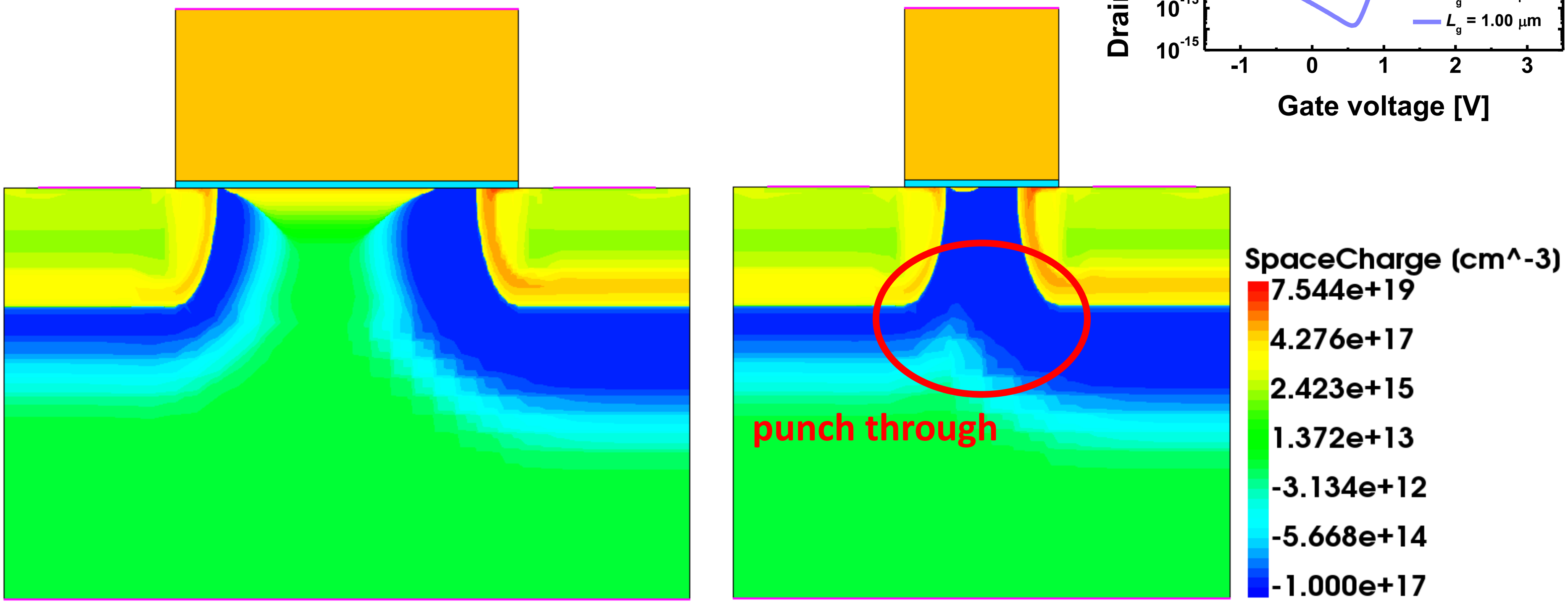




Punch through

- ✓ V_{GS} = -1.0 V, V_{DS} = 5.0 V

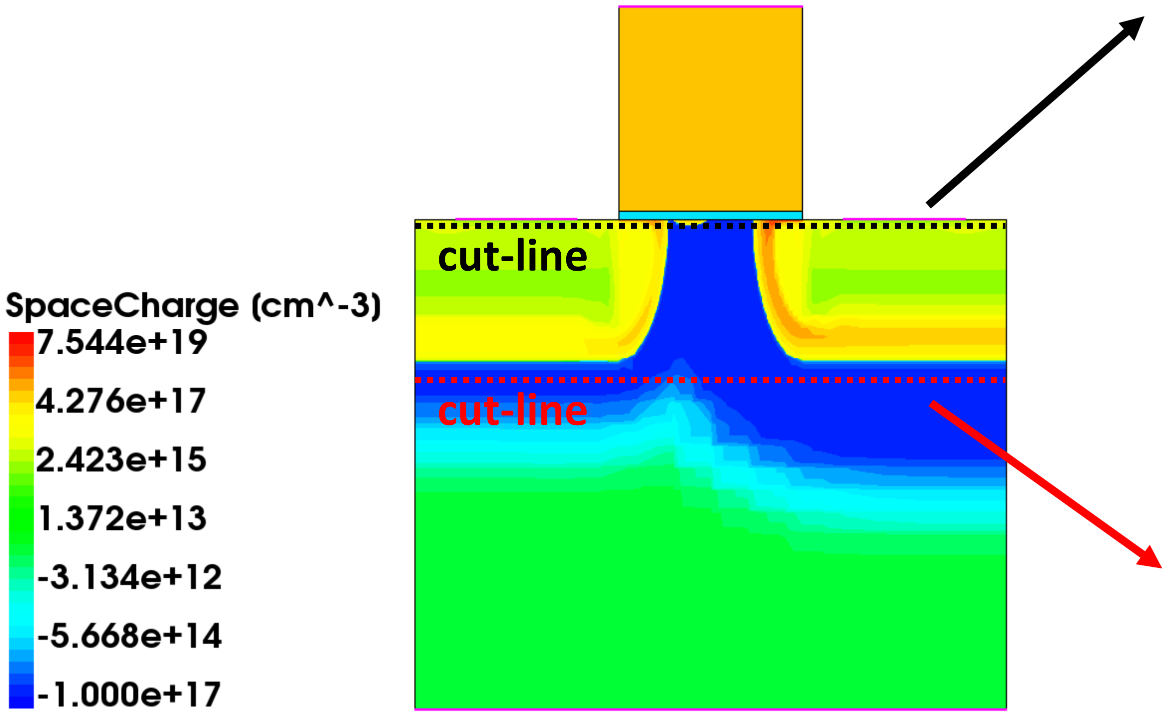


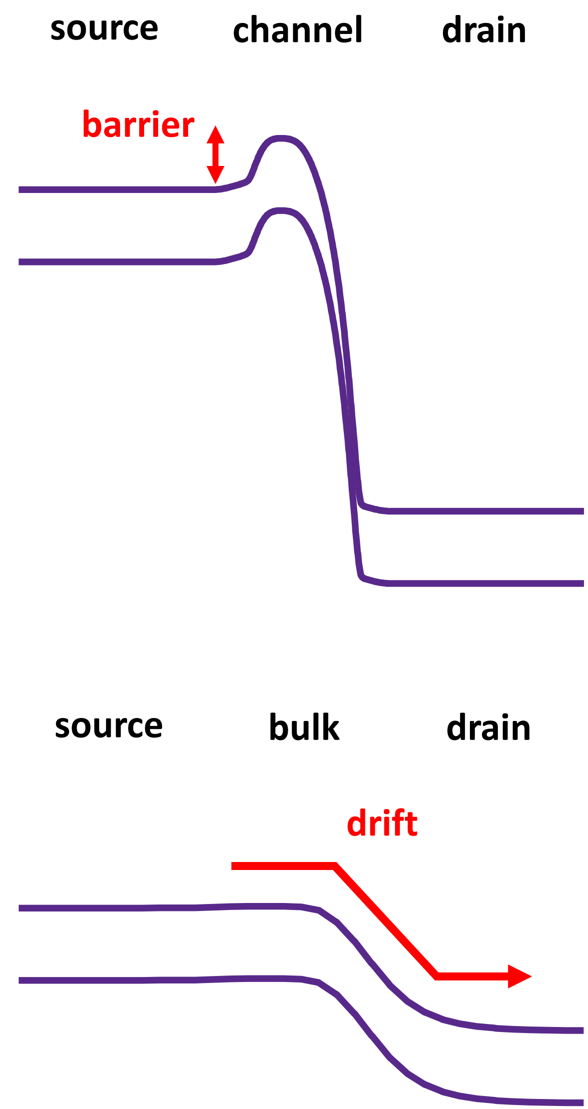


Punch through

✓ V_{GS} = -1.0 V, V_{DS} = 5.0 V

L_g = 0.45 μm

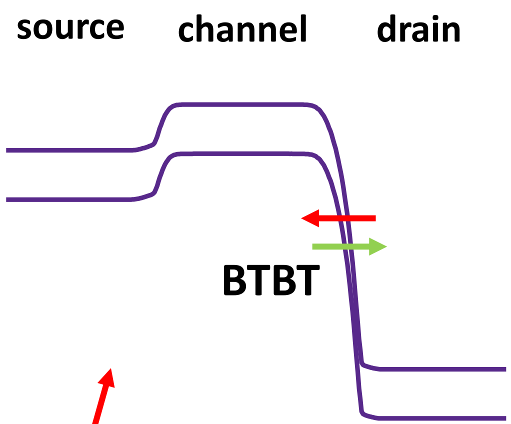


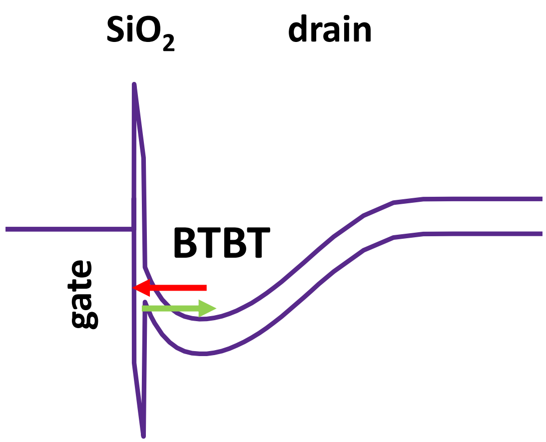


GIDL

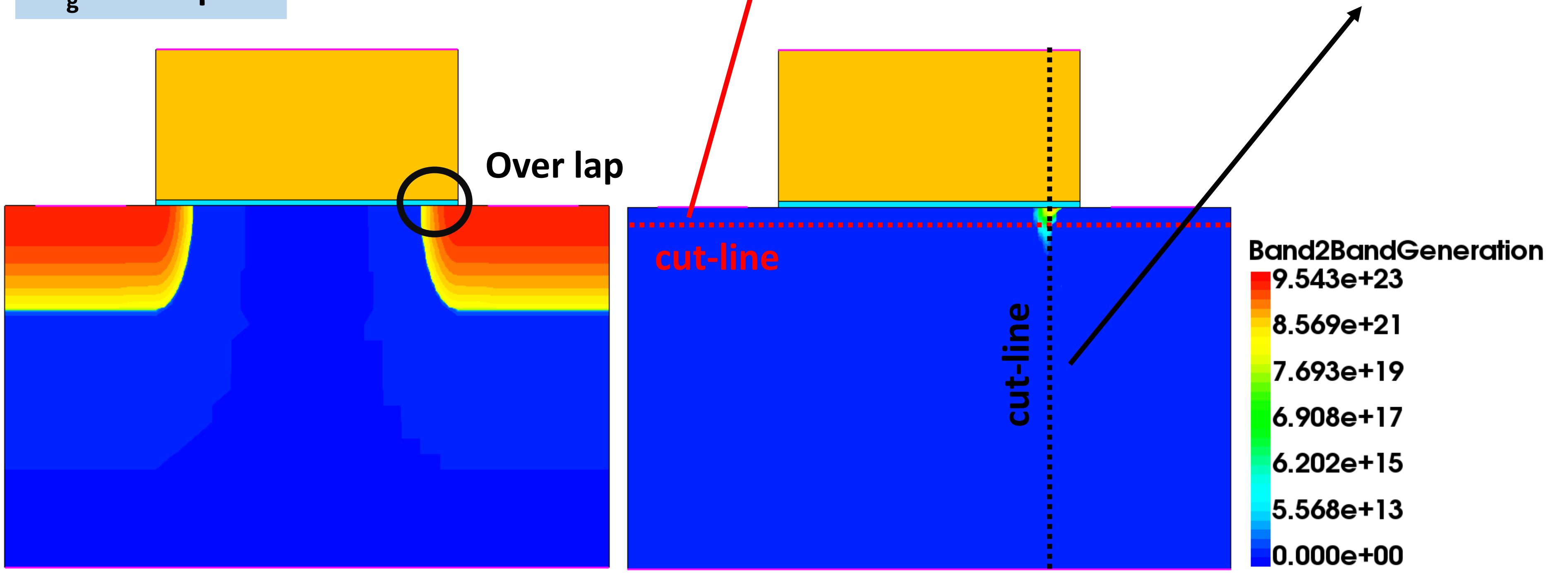
- ✓ V_{GS} = -1.0 V, V_{DS} = 5.0 V
- ❖ Physics의 recombination (Band2Band (E1)) 적용시 BTBT구현 가능

* band-to-band-tunneling (BTBT)





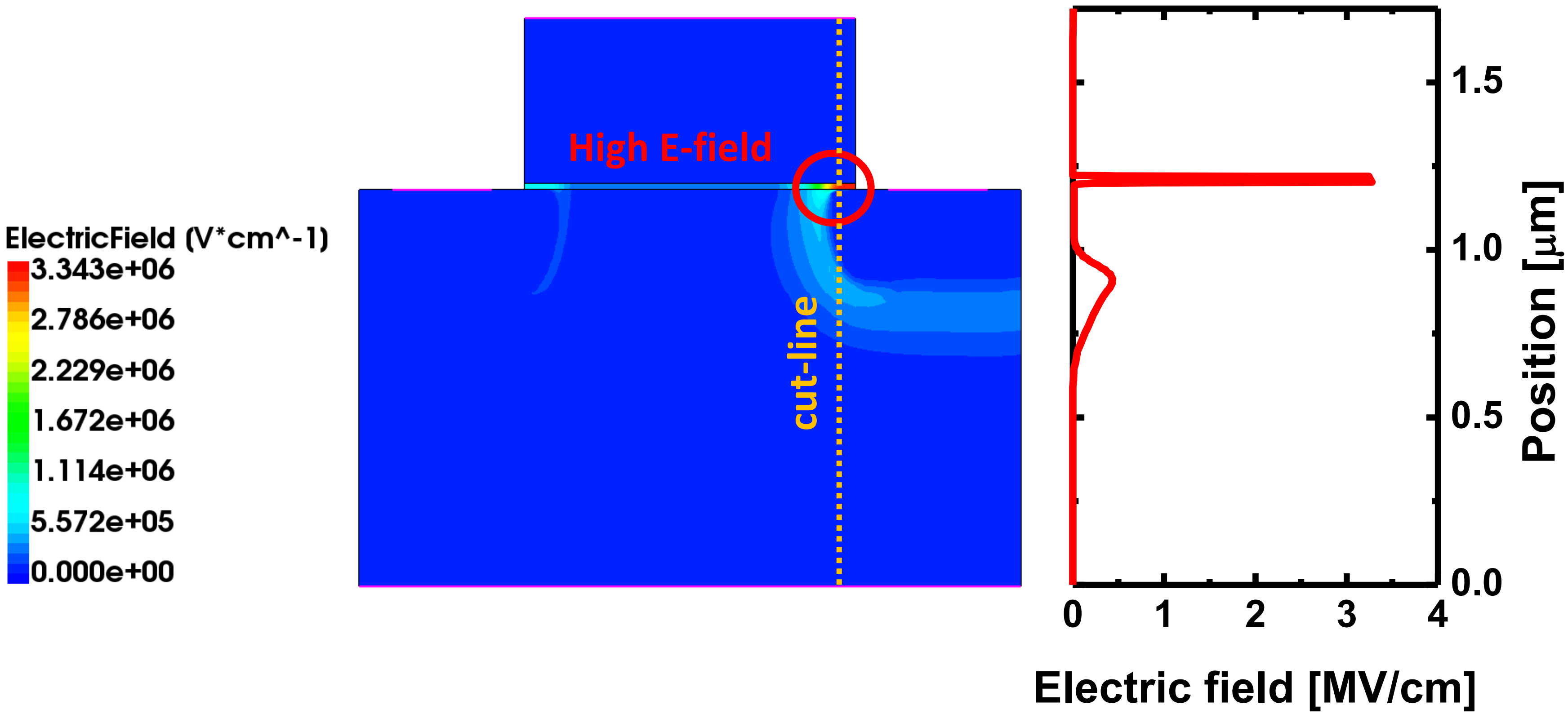
L_g = 1.00 μm



GIDL

✓ V_{GS} = -1.0 V, V_{DS} = 5.0 V

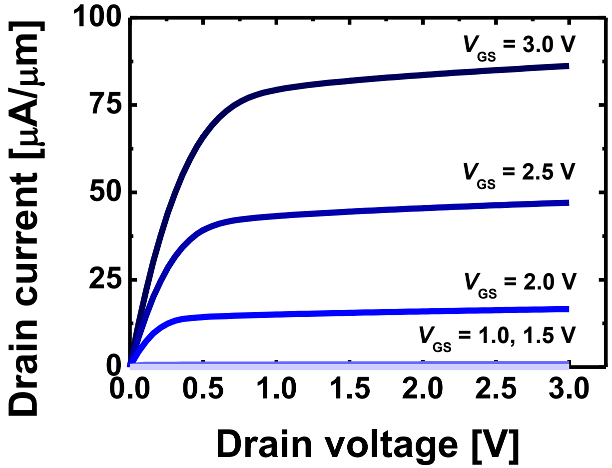
L_g = 1.00 µm



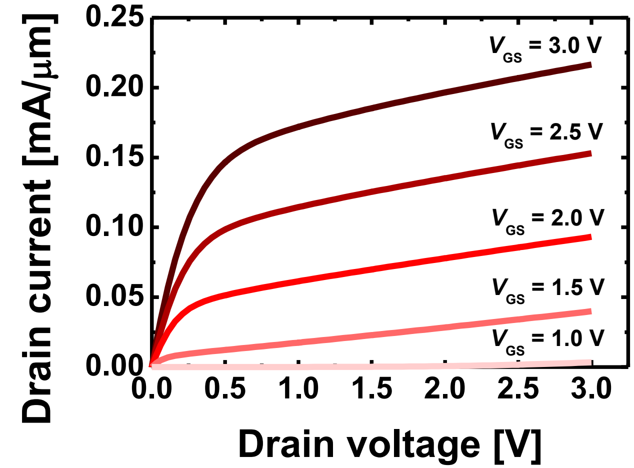
Content

Summary

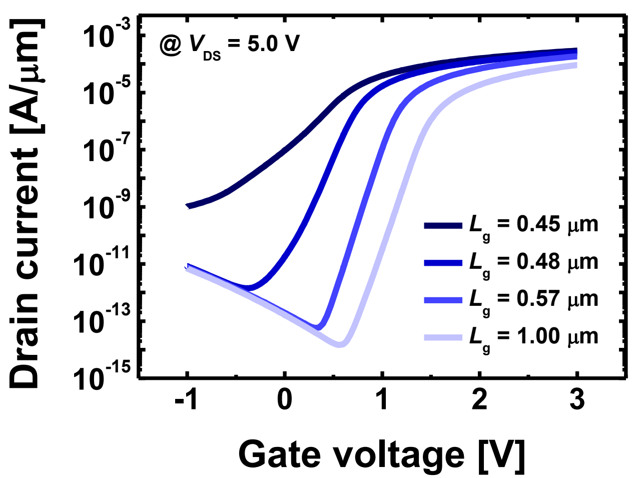
I_D-V_D



Channel length decrease



I_D-V_G



L_g [μm]	V_{th} [V]	SS [mV/dec]	DIBL [mV/V]
1.00	1.37	111.63	4.49
0.57	1.04	108.99	53.27
0.48	0.58	131.83	124.29
0.45	0.13	337.94	202.86